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NITI Aayog

DRAFT



SECTOR AT A GLANCE

Agriculture and allied sectors in India

AN OVERVIEW

Sector at a Glance

Agriculture and Allied Sectors in India

An Overview

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

India's agriculture and allied sectors play a pivotal role in the socioeconomic development and progress of the nation. Although these sectors contribute to just 19% of the national income, they employ almost half of the country's workforce with more than 70% of rural households in India depending on them for their livelihood.¹ Crops (54%) and livestock (30%) are the major drivers of the sectors followed by forestry (9%) and fisheries (7%). Exports of agricultural commodities constituted 12% of the total national exports while imports made up 5% of the total national imports in 2021-22.

Enhancing farmers' incomes and improving the nutritional security and climate resilience of Indian agriculture and allied sectors via soil health management, water-use efficiency, research, innovation and technology are the key policy focus areas in these sectors. These are reflected in the numerous initiatives undertaken by the Government – for e.g schemes such as PM-KISAN, Soil Health Card, Pradhan Mantri Krishi Sinchayee Yojana, Agri Infrastructure Fund, and Livestock Health and Disease Control; National Fisheries Policy; and National Agriculture Market portal, etc., – to improve the productivity and sustainability of these sectors. The Government is also focused on enabling sustainable agriculture and developing climate-resilient technologies, which are central to achieving the Sustainable Development Goal (SDG-2) on zero hunger before 2050.²

Economic Value

The Indian agriculture and allied sectors were valued at **INR 41 lakh crores** and contributed to **19% of India's Gross Value Added (GVA)³ in 2021-22**. It was largely driven by the sub-sectors of **crops (54%) and livestock (30%)** followed by forestry (9%) and fisheries (7%). In recent years, India has rapidly emerged as the net exporter of agricultural products growing at an average annual rate of 12% during the last five years. The sectors grew at an average annual growth rate of 4.5% over a five-year period from 2017 to 2022.

Figure 1: Overview of Sectoral Composition and Economic Contribution of Agriculture and Allied Sectors in India

Agriculture and Allied Sectoral Composition and Contribution to India's GVA

10.4%
Crops (Field crops*
and Horticulture)

5.7%
Livestock and Dairy

1.5%
Forestry

1.3%
Fisheries and
Aquaculture

= 19%
Agriculture
and Allied

*Field crops include food crops and commercial crops

Key Data Points of the Agriculture and Allied Sectors

INR 41 Lakh Cr
GVA in 2021-22[^]

4.5%
average annual growth
rate in 2017-2022

45%
labour force employed
in 2021-22

12%
contribution to Indian
exports in 2021-22

Economic Value

Employment

International Trade

Note: [^]The agriculture and allied sectors experienced a nominal annual growth of 11.2% in 2022-23 (compared to 3.3% in 2011-12 constant prices) taking the nominal GVA to INR 45 lakh crores

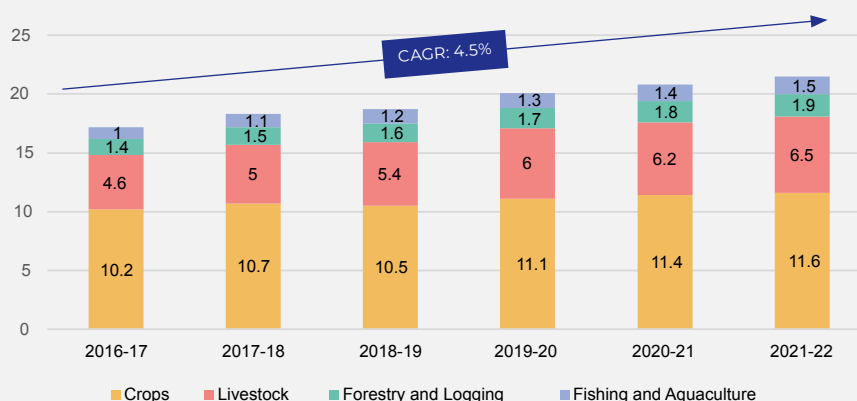
Sources: Agriculture Statistics at a Glance 2022, Periodic Labour Force Survey 2021-22

1 Ministry of Finance. (2021). *Economic Survey 2020-21*.

2 Indian Council of Agricultural Research. (2015). *Vision 2050*.

3 Ministry of Agriculture and Farmers Welfare. (2023). *Agriculture Statistics at a Glance 2022*.

Figure 2: Gross Value Added by Agriculture and Allied Sectors at Constant (2011-12) Prices (INR Lakh Crores) from 2017-2022



Note: Crops include field crops (food crops, commercial crops) and horticulture; livestock includes milk, meat, eggs and wool

Source: Agriculture Statistics at a Glance 2022

Employment and Income

The agriculture and allied sectors employ roughly **45% of the workforce (25 crore persons)** in the country.⁴ The majority of farmers in India are small and marginal farmers (86%) who rely on daily wages to supplement farm income; these wages constitute 30-60% of the total annual incomes of their households.⁵

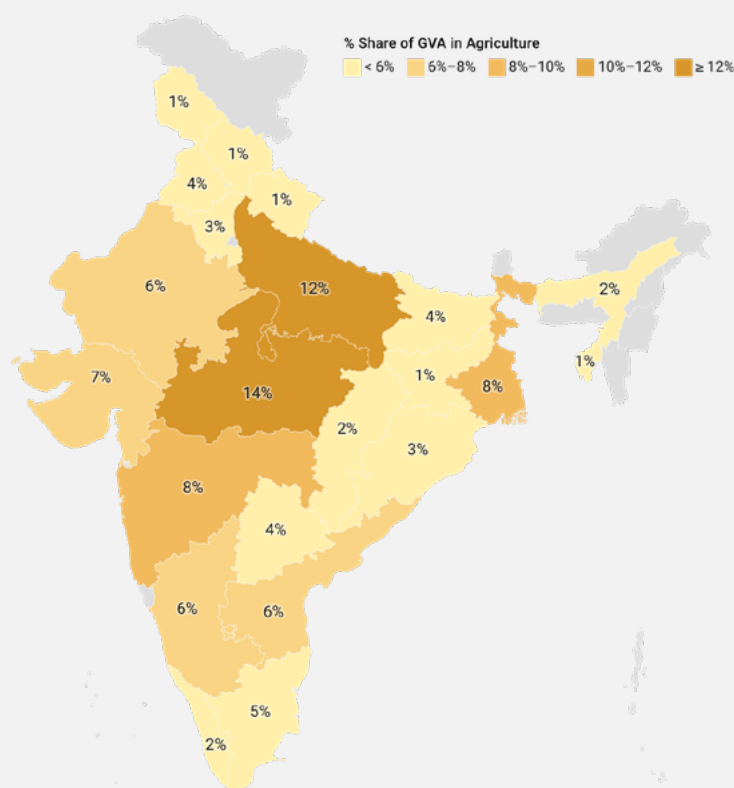
International Trade

Agriculture exports, valued at INR 3.7 lakh crores in 2021-22, **contributed to 12% of total national exports**. Similarly, the share of **agriculture imports**, valued at INR 2.3 lakh crores, to overall national imports is 5.1%.³

Value of Output by States

Despite being a mainstay of the economy and leading generator of employment, **~76% of India's total agricultural value** is concentrated in ten states with Madhya Pradesh contributing to 14% of the total GVA from agriculture.⁶

Figure 3: Gross State Value Added by Agriculture and Allied Sectors in 2019-20.⁷



Source: RBI Handbook of Statistics on Indian States

⁴ Ministry of Statistics and Programme Implementation. (2023). *Periodic Labour Force Survey (PLFS) 2021-2022*.

⁵ Ministry of Statistics and Programme Implementation. (2021). *Situation Assessment Survey of Agricultural Households and Land and Livestock Holdings of Households in Rural India, 2019*.

⁶ Reserve Bank of India (n.d.). *Handbook of Statistics on Indian States – Table 28*.

⁷ The Gross State Value Added was calculated for the year 2019-20 as there were missing values present in the latest available data (2021-22).

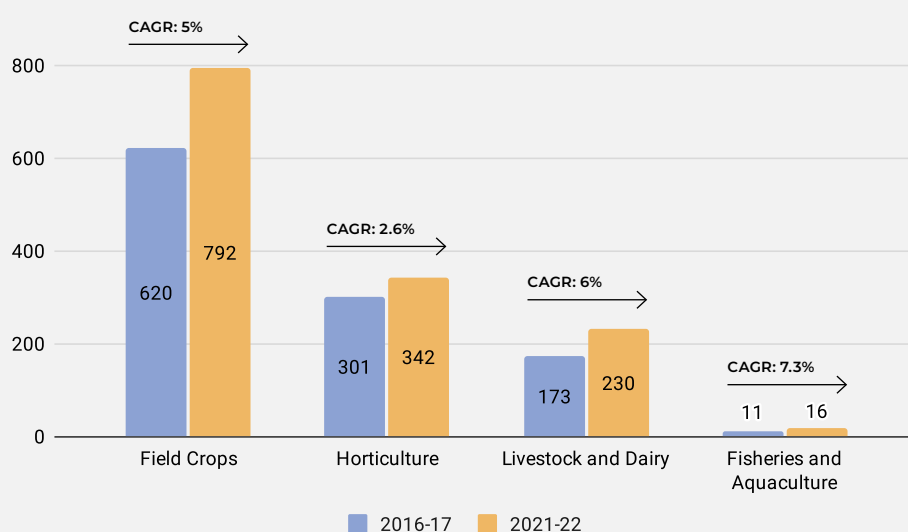
Sub-sectoral Trends

This section focuses on trends in production and international trade⁸ of the following agriculture and allied sub- sectors:

1. Field Crops
2. Horticulture
3. Livestock and Dairy
4. Fisheries and Aquaculture
5. Forestry

There has been a strong positive trend in the annual growth rate of production across all sub-sectors, especially in livestock and dairy, and fisheries and aquaculture.

Figure 4: Sub-sectoral production trends from 2017 to 2022 [in million metric tonnes (MMT)]



Note: The livestock and dairy sub-sector includes milk and meat production. Forestry products include industrial wood, fuelwood, and Non-Timber Forest Products (NTFPs) such as fruits, vegetables, medicinal plants, gums, resins, bamboo, ratans and a variety of other palms and grasses. The production trend is not included in the chart as there is no common metric that captures the volume information of the sectors.

Source: Agriculture Statistics at a Glance 2022

a. Field Crops

Field crops are further classified into food crops and commercial crops, and oilseeds.

i. Food Crops

Food crops include **cereals (rice and wheat), and nutri and coarse cereals and pulses**. They have been the key focus of the Government of India since the Green Revolution which inspired the nation's objective of ensuring its food security.

⁸ Agriculture Statistics at a Glance 2022 and Agricultural and Processed Food Products Export Development Authority (APEDA) are the primary sources of data used in this document for production and international trade statistics respectively for the agriculture and allied sub-sectors

Production⁹

316 MMT

Total production of food crops in 2021-22

2.8%

CAGR (2017-22)

75%

of food crops are rice and wheat

25%

of food crops production are pulses and coarse cereals

More than half (52%) of the food crops produced in India are from the states of Uttar Pradesh (17.8%), Madhya Pradesh (12.4%), Punjab (9%), Rajasthan (6.7%) and West Bengal (6.5%).

Trade

Among the food crops, rice and pulses are the major commodities exported and imported respectively.

Rice (including Basmati and Non-Basmati varieties) contributed to 21% (in value terms) of total agriculture principal commodity exports in 2023 with a five-year average annual growth rate of 7% for Basmati rice and 17% for non-Basmati rice.¹⁰ **Pulses** are the third largest principal agricultural commodity imported in 2023 valued at INR 15,780 crores (6% of total agricultural imports).¹¹

Crop Diversification

Diversification among food crops is practised to increase food security, reduce undernutrition, and increase climate and pest resilience. Millet cultivation is being encouraged to ensure food, health, nutrition and economic security as well as soil health maintenance. States like Odisha, Chhattisgarh and Tamil Nadu have their own Millet Mission programmes. States such as Andhra Pradesh, Madhya Pradesh, Haryana, Nagaland and Uttarakhand are introducing policies focused on specific millets.

India is the leading producer of millets in the world with an area of 15.5 million hectares under cultivation for the same in 2021-22.

ii. Commercial Crops

The commercial crops cultivated widely in India are **cotton, sugarcane, jute, copra¹²**, tea, coffee and tobacco unmanufactured.

Production¹³

439 MMT

Total production of cotton, sugarcane, and jute in 2021-22

-0.5%

CAGR (2017-22)

1st

India's rank in jute production

2nd

India's rank in cotton & sugar production

Sugarcane constitutes 95% of the total commercial crops produced and has the highest average annual growth in production among the commercial crops at 2.6%.

9 The Ministry of Agriculture & Farmers Welfare (MoAFW) released the third advance estimates for the 2022-23 data on production in May 2023. According to those estimates, total production of food crops in 2022-23 was 330.5 MMT at a CAGR of 3% since 2017-18.

10 [Agricultural and Processed Food Products Export Development Authority. \(n.d.\). Agri Exchange – India Export of Principal Commodities \(All Agri\).](#)

11 [Agricultural and Processed Food Products Export Development Authority. \(n.d.\). Agri Exchange – India Import of Principal Commodities \(All Agri\).](#)

12 Copra, originating from coconuts, is an important commercial product in India. However, coconut has traditionally been considered a plantation crop falling under horticulture, leading to ambiguity in the categorisation of copra. For the purpose of this Sector-at-a-Glance, coconut has been included under horticulture. Further details on coconut and copra can be found in both Horticulture and Commercial Crops Sectors-at-a-Glance.

13 According to MoAFW Third Advance Estimates, total production of commercial crops (excluding coconut) in 2022-23 was 538 MMT at a CAGR of 5% since 2017-18

Trade

Among the commercial crops, sugar is the most exported commodity. It constitutes 11% (in value terms) of total agriculture principal commodities exports in 2023 growing annually at 55% since 2018-19. Raw cotton valued at INR 11,506 crores (4% of total agricultural imports) is the most imported commodity among commercial crops.

Sustainability and Product Diversification

Great emphasis has been laid on adopting sustainable cultivation practices like drip irrigation and intercropping for crops like sugarcane and cotton to reduce the rate of groundwater extraction. Product diversification is crucial for long-term economic viability of coconut and jute which can be used to create a diverse range of products such as jelly, vinegar (extracted from coconut) and woven and non-woven geotextiles (extracted from jute). This has been made possible through the modernisation of mills which now use high-quality processing infrastructure.¹⁴

iii. Oilseeds

The nine annual oilseeds grown in India include **seven edible oilseeds viz., groundnut, rapeseed-mustard, soybean, sunflower, sesame, safflower and niger, and two non-edible varieties viz., castor and linseed.**

Production¹⁵

38 MMT

Total production of oilseeds in 2021-22

3.8%

CAGR (2017-22)

1st

India's rank in groundnut production

30.5% of the total oilseed production in the country is contributed by Rajasthan (8.4%), Madhya Pradesh (8%), Gujarat (7%), Maharashtra (6%) and Haryana (1.4%).

Trade

Among the oilseeds, vegetable oils are the largest principal agricultural commodity imported in 2023 valued at INR 1.6 lakh crores (61% of total agricultural exports). Oil meals, castor oil and groundnut account for 7% (in value terms) of total agriculture principal commodities exports in 2023.

Although India is the fourth largest edible oil economy globally, contributing to about 10% of the world's oilseed production, it is one of the largest importers of edible oils in the world.¹⁶

c. Horticulture

The horticulture sub-sector is composed of products such as **fruits, vegetables, plantation crops, spices, floriculture, aromatics and medicinal plants.** It plays an important role in the development of the rural economy by creating additional livelihood opportunities, especially for women farmers.

¹⁴ Shahinur et al. (2022). *Current Development and Future Perspective on Natural Jute Fibers and Their Biocomposites*.

¹⁵ According to MoAFW Third Advance Estimates, total production of oilseeds in 2022-23 was 41 MMT at a CAGR of 5.5% since 2017-18

¹⁶ Teja et al. (2017). *Performance of oilseeds in India – a temporal analysis*.

Production

342 MMT

Total horticultural production in 2021-22

2.6%

CAGR (2017-22)

1st

India's rank in the production of mango, banana, coconut, papaya, pomogranate, ginger & okra

Almost 50% of the horticulture crops produced in India are from the states of Uttar Pradesh (12%), Madhya Pradesh (10%), West Bengal (10%), Maharashtra (9%) and Andhra Pradesh (8%).

Trade

Among horticulture crops, spices and fresh fruits are the main commodities that are exported and imported respectively. The export of spices valued at INR 30,433 crores in 2022-23 (7% of total agriculture exports) has grown at an average annual rate of 9% since 2017-18. **Fresh fruits** are the second largest imported agricultural commodity constituting 7% of the total agricultural imports (in value terms) in 2022-23 with a CAGR of 10% since 2017-18.

One-third of the products under the One District One Product (ODOP) scheme are horticulture crops. The scheme focuses on branding and promoting one product from each district of the country to enable holistic socioeconomic growth across all regions.

d. Livestock and Dairy

The livestock and dairy sub-sector, composed of products such as **milk, meat, eggs, and wool**, is instrumental in providing livelihood opportunities to small and marginal farmers in India.

i. Dairy

Production

221 MMT

Total milk production in 2021-22

6%

CAGR (2016-22)

1st

India's rank in milk production

24%

India's contribution to global milk dairy production in 2021-22

53% of the national milk production is from the states of Rajasthan (15%), Uttar Pradesh (15%), Madhya Pradesh (8.6%), Gujarat (7.6%) and Andhra Pradesh (7%)

Trade

Dairy products constitute a minor share in the international trade market. Export of dairy products valued at INR 4,709 crores (1% of total agricultural exports¹⁷) grew at a CAGR of 19% from 2017-18 to 2022-23. Import of dairy products valued at INR 444 crores (0.2% of total agricultural imports¹⁸) grew at a CAGR of 7% during the same period.

The Indian dairy market size is expected to double to INR 30 lakh crore by 2027 due to increase in production of milk and dairy products as well as value appreciation.¹⁹

¹⁷ Agricultural exports include exports from allied sectors such as livestock, dairy, fisheries and forestry.

¹⁸ Agricultural exports include exports from allied sectors such as livestock, dairy, fisheries and forestry.

¹⁹ Agri Exchange. (2022). Size of Indian dairy market to jump over 2-fold to Rs 30 trn by 2027: NDDB.

ii. Meat

Production

9 MMT

Total meat production
in 2021-22

4.6%

CAGR (2017-22)

51%

of meat production is driven
by poultry

Half of the national meat production is driven by poultry followed by meat from buffalo, goats, sheep, pigs and cattle. 58% of total meat production in the country is from Maharashtra (12.3%), Uttar Pradesh (12%), West Bengal (11.6%), Andhra Pradesh (11%) and Telangana (10.8%).

Trade

Buffalo meat is the country's most exported livestock product. It is valued at INR 25,640 crores (6% of total agriculture exports) in 2022-23 and had almost nil growth since 2017-18. The import of poultry products, and sheep, goat and other meat accounts for less than 0.1% (in value terms) of total agricultural imports in 2023.

e. Fisheries and Aquaculture

The fisheries and aquaculture sub-sector is composed of inland and marine fisheries which provide livelihood to a population of over 2.8 crores.²⁰ It plays a major role in ensuring food security and provides a sustainable source of protein to the Indian population.

Production

16 MMT

Total fish production in 2021-22

7.3%

CAGR (2017-22)

8%

India's share in global fish
production

Inland fisheries account for 75% of the total fish production in India while marine fisheries account for the remaining 25%. More than half (54%) of the fish production in India is from the states of Andhra Pradesh (30%), West Bengal (11%), Karnataka (7%) and Orissa (6%).

Trade

Marine products, valued at ~INR 65,000 crores, account for 15% of total agricultural exports. These have grown at a CAGR of 6% from 2017-18 to 2022-23. The import of marine products accounts for ~1% (in value terms) of total agricultural imports in 2023.

Globally, India is the third largest fish and aquaculture-producing country accounting for 16% of the total inland and 5% of total marine fish production respectively.²⁰

20 Ministry of Fisheries, Animal Husbandry & Dairying. (2022). *Handbook on Fisheries Statistics 2022*.

f. Forestry

The forestry sub-sector is composed of **major forest products like industrial wood and fuelwood and minor forest products like bamboo, lac, resin, gum etc.** About 25% of the country's forest products are from Maharashtra (15%) and Rajasthan (10%) followed by Uttar Pradesh, Madhya Pradesh, Punjab and Odisha.

Production

402 MMT

Estimated green bamboo culms in 2021

85.3 MMT

Total fuelwood extraction from forests in 2019

1st

India's rank in guar gum production

Trade

Forest products like shellac and guar gum constitute a minor share of the international trade market. Exports of **guar gum** valued at INR 4,945 crores (1% of total agricultural exports) and **shellac** valued at INR 851 crores grew at a CAGR of 3% and 24% respectively from 2017-18 to 2022-23. NTFPs constitute about 70% of the earnings generated from forest-related exports, amounting to USD 500 million. Imports of shellac and guar gum account for 0.1% (in value terms) of total agricultural imports in 2023.

Agroforestry, a land use system integrating the growth of trees and agricultural products, currently **fulfils approximately 72% of the total fuelwood demand** and also addresses rural household needs such as food, fruits, fodder, fibre, fuel, medicine, etc.²¹

Opportunities & challenges

Opportunities



Growing population and demand for food:

- India's population is expected to grow to 165 crores by 2050² which calls for increasing agricultural productivity and improving the supply chain to meet the subsequent increase in food demand.



Post-harvest management infrastructure:

- Various Government of India schemes aims to modernize the agri-value chain by promoting farmer producer organisations, agri-logistics, processing facilities and professional management of such marketing.²²
- The Indian processed food market is expected to grow to INR 34 lakh crores by 2025, a 78% increase from 2020 due to government initiatives such as planned infrastructure worth INR 80 lakh crores, the Pradhan Mantri Kisan Sampada Yojna, Gati Shakti, Animal Husbandry Infrastructure Development Fund (AHIDF), Fisheries and Aquaculture Infrastructure Development Fund (FIDF), etc.²³

²¹ Sahoo et al. (2019). *Multifunctional agroforestry systems in India for livelihoods*.

²² NITI Aayog. (2018). *Strategy for New India @ 75*.

²³ India Brand Equity Foundation (2023). *Agriculture in India: Information About Indian Agriculture & Its Importance*. Retrieved 23 May 2023.



Innovation and technology:

- The Indian agri-tech ecosystem has the potential to boost farmers' incomes by 25- 35% and add INR 7.8 lakh crores to the economy by 2030.²⁴
- In 2022, the Indian Council of Agricultural Research developed a roadmap for the next ten years to secure nutrition and income security and promote farmers' use of new technology to create improved seed varieties and new breeds.²⁵
- Over 7,000 farmers in India are using artificial intelligence technologies for crop health monitoring, soil testing and quality control via Artificial Intelligence for Agriculture Innovation initiative by the World Economic Forum²⁶



Major focus on allied sectors:

- The Government of India announced the highest ever investment – INR 30,572 crores – towards the fisheries sub-sector for the years 2015-25.²⁷
- Household consumption of poultry and other meat is expected to grow from 12% to 24% by 2030.²⁸
- Globally, India ranked third with respect to the net gain in average annual forest area between 2010-2020²⁹ and has the potential to further promote sustainable land use practices such as agroforestry.

Challenges

Poor access to raw materials, technology, modern infrastructure, credit, and markets, and the overexploitation of natural resources are some of the major challenges faced by the agricultural and allied sectors. These challenges lead to reduced productivity, low farmers' incomes and poor climate resilience. The major challenges can be classified into four categories: sectoral, input-level, production and post-production challenges.



Sectoral challenges

- **Low penetration of agri-tech:** Penetration of the Indian agri-tech market (~INR 2 L Cr) remains critically low at less than 1%.³⁰
- **Access to finance:** Informal sources of credit constitute 40% of loans accessed by cultivator households.³¹
- **Implementation of risk mitigation strategies:** Agricultural insurance schemes are hard to implement given the vastly fragmented landholding and lack of data.
- **Low productivity:** Productivity remains low due to slow adoption of modern technology, mechanisation and agricultural research and development.
- **Climate change:** Climate change has a negative impact on food security and rural livelihoods.
- **Loss of crop diversity:** Unsuitable crop patterns and allocation of acreage lead to a loss of crop diversity.

24 McKinsey & Company (2023). *How agtech is poised to transform India into a farming powerhouse*.

25 Press Information Bureau. (2022). *Research and Development in the Agriculture Sector*.

26 World Economic Forum. (2023). *AI for agriculture: How Indian farmers are harnessing emerging technologies to sustainably increase productivity*.

27 Department of Fisheries. (2021). *Transforming Fisheries Pradhan Mantri Matsya Sampada Yojana*.

28 Department of Animal Husbandry and Dairying. (2022). *National Action Plan for Egg & Poultry For Doubling Farmers' Income by 2022 (MoAFW, 2022)*.

29 Ministry of Finance. (2023). *Economic Survey 2022-23*.

30 Ernst & Young. (2020). *AgriTech - towards transforming Indian agriculture*.

31 National Bank for Agriculture and Rural Development. (2021). *Trends and Patterns in Agriculture Credit in India: A District Level Analysis of Uttar Pradesh*.

 Input-level challenges	• Small holdings: 86% of Indian farmers have small land holdings (less than two hectares), leading to low incomes.³² • Deteriorating soil and water conditions: 5.3 Bn T of soil is eroded annually (16.4 T/ha)³³ and 87% of the total annual groundwater extraction is for irrigation.³⁴ • Improper use of agricultural inputs: Poor quality seed, planting material and the overuse of fertilizers and pesticides affect the overall crop productivity and soil health. • Issues with feed and fodder: There is a shortage of good quality feed and fodder leading to high costs; over 60% of the operational expenditure in aquaculture comprises of feed expenses.³⁵ • Limited extension services: Productivity is diminished due to low access to knowledge and training on modern technology and best practices for farmers.
 Production-level challenges	• Outbreak of pests and diseases: In India, pests and diseases cause over INR 290 billion per annum losses of crops;³⁶ the health and productivity of livestock are affected by frequent outbreak of diseases like Foot and Mouth Diseases, Black Quarter infection, Influenza etc. • Slow adoption of new production techniques and technologies: The uptake of modern technology for crop production forecasting and sustainable marine production techniques (e.g. mariculture and mechanised boats with the latest GPS and satellite tracking systems) is yet to gain momentum. • Inadequate infrastructure facilities: Agriculture infrastructure such as farm machinery hubs and public irrigation need upgradation; slaughterhouses in India lack basic facilities for hygienic slaughter of animals for consumption. • Low adoption rate of agroforestry practices: A shortage of quality planting material, research, market infrastructure, and streamlined legislation with respect to tree-felling, wood transportation, and processing has affected the growth of agroforestry.
 Post-production-level challenges	• Inadequate post-harvest infrastructure: The lack of effective cold-chain infrastructure has led to high post-harvest losses (mainly of perishables) estimated at INR 92.6 Cr annually. Food transport is affected by logistical barriers (for instance, national highways account for 40% of all cargo).²⁹ • Inadequate marketing infrastructure: Retail shops lack the basic facilities to handle meat in a hygienic manner, which influences the purchasing behaviour of consumers; upgradation of rural markets required. • Inefficient markets: Only 14% of farmers have registered on the National Agriculture Market portal (eNAM), of which 51% have not received benefits.³⁷ • Limited alignment with global standards and certification: India has only one fishery resource certified by the international agency, the Marine Stewardship Council, and faces challenges with the certification of organic products.

Government Interventions

Ministries and Departments

The agriculture and allied sectors in India have two core ministries – the Ministry of Agriculture and Farmers Welfare and the Ministry of Fisheries, Animal Husbandry and Dairying – and four major departments namely the Department of Agriculture and Farmers Welfare, the Department of Agriculture Research and Education, the Department of Animal Husbandry and Dairying and the Department of Fisheries.

³² Ministry Of Agriculture & Farmers Welfare. (2019). Agriculture Census 2015-16.

³³ Saroha. (2017). Soil Erosion: Causes, Extent and Management in India.

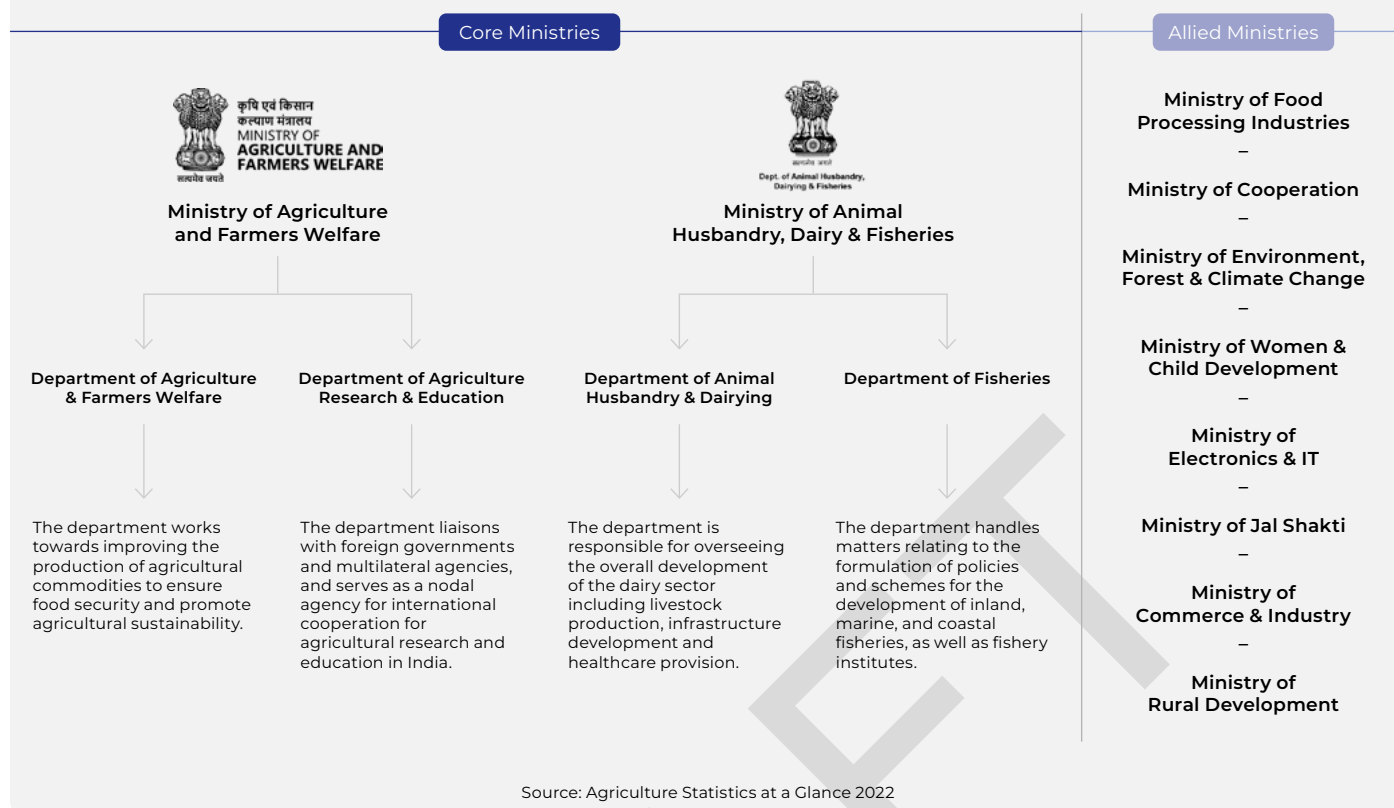
³⁴ Ministry of Jal Shakti. (2022). National Compilation on Dynamic Ground Water Resources Of India, 2022 (pp. 73).

³⁵ Paul, B.N. et al. (2017). Application and Practices of Fish Feed in Aquaculture.

³⁶ Shukla, S. et al. (2022). Challenges Faced by Farmers in Crops Production Due to Fungal Pathogens and Their Effect on Indian Economy.

³⁷ Das, A. et al. (2020). Indian e-commerce scenario - An overview.

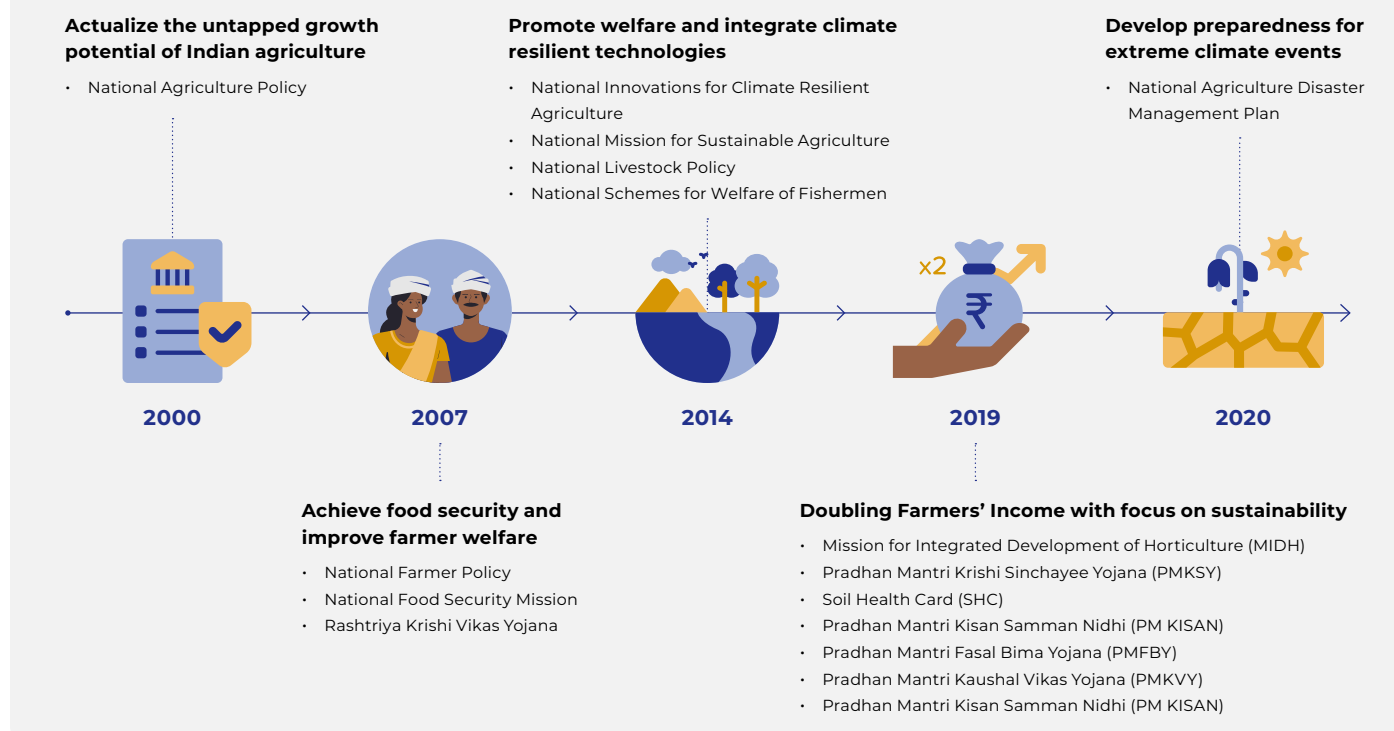
Figure 5: Key Ministries and Departments of the Agriculture and Allied Sectors



Policy Evolution over the Decades

There has been a shift in India's food and farming system from focusing on mere production to enhancing farmers' incomes and ecological security.³⁸

Figure 6: Evolution of Policy Vision across Agriculture, Livestocks and Fisheries from 2000-2020



38 Chand et al. (2021). *Indian Agriculture towards 2030*.

Policies, schemes, and initiatives

The major policies, schemes³⁹ and initiatives introduced by the Government of India for the agriculture and allied sectors have been classified into four categories: sectoral, input-level, production and post-production.

Figure 7: Sectoral Interventions



SECTORAL INTERVENTIONS

Pradhan Mantri Kisan Samman Nidhi	Agricultural Education & Research	Green Revolution – Krishonnati Yojana*
Crop Insurance Scheme - PMFBY	Market Intervention Scheme and Price Support Scheme	Modified Interest Subvention Scheme (MISS)
Pradhan Mantri Matsya Sampada Yojana (PMMSY)	Formation and Promotion of 10,000 Farmer Producer Organisations (FPOs)	Rashtriya Krishi Vikas Yojana (RKVY)
National Livestock Mission	Livestock Census and Integrated Sample Survey	Pradhan Mantri Annadata Aay Sanrakshan Yojna (PM-AASHA)

*An umbrella scheme covering 11 schemes and missions such as the Mission for Integrated Development of Horticulture (MIDH), National Food Security Mission (NFSM), Sub-Mission on Agriculture Extension (SMAE), Sub-Mission on Seeds and Planting Material (SMSP), Sub-Mission on Agricultural Mechanisation (SMAM), Integrated Scheme on Agricultural Marketing (ISAM) and others.

Figure 8: Input-level, Production-level and Post-production-level Interventions



INPUT-LEVEL INTERVENTIONS

Soil Health Card	Pradhan Mantri Krishi Sinchayee Yojana	Rashtriya Gokul Mission
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PRODUCTION-LEVEL INTERVENTIONS

Livestock Health and Disease Control Scheme	National Fisheries Policy 2020	National Agroforestry Policy 2014
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POST PRODUCTION-LEVEL INTERVENTIONS

Agri Infrastructure Fund	National Agriculture Market	Production Linked Incentive Scheme for Food Processing Industry
National Programme for Dairy Development	Dairy Processing & Infrastructure Development Fund	

³⁹ Only the schemes that received an average budgetary allocation of INR 500 crores and above by the Government of India during 2019-20 to 2022-23 have been included.

Way Forward

The agriculture and allied sectors are the largest source of livelihoods in India with a budget allocation of INR 1.25 lakh crores in 2023-24.⁴⁰ The Government of India has increased its focus on making the country agriculturally self-reliant and improving the lives and livelihoods of small and marginal farmers. Looking ahead, there are four key themes that could shape the Indian agricultural sector.

 Enhancing and protecting smallholder farmer income	Improving farm and non-farm productivity, and market access - Emphasising on yield gap minimisation by incorporating new technologies, precision farming and R&D in agriculture - Strengthening breed improvement and last-mile delivery of AI services in India. - Enabling role of Farmer Producer Organisations (FPOs) for better market access and opportunities for small farmers	Improving income outcomes for farmers PM-KISAN, a DBT scheme, addresses liquidity constraints of farmers in buying agri inputs - Increasing adoption of Kisan Credit Card scheme to avail timely credit from banks for farm expenses - Ensuring crop and livestock insurance availability for agricultural and allied farmers - Strengthening farmers' efforts and risk mitigation via RKVY-RAFTAAR scheme
 Incorporating sustainability	Optimizing soil and water management - Integrating nutrient management in soil with organic and inorganic fertilizers - Assessing soil health indicators using the Soil Health Card scheme by farmers - Adopting water-smart technologies like micro irrigation (Pradhan Mantri Krishi Sinchayee Yojana) with System of Rice Intensification (SRI) and precision farming - Promoting nation-wide adoption of natural farming (low cost, chemical-free inputs) via Bhartiya Prakritik Krishi Padhati scheme	Increasing climate resilience - Mainstreaming of climate-resilient crops such as millets which grow well in relatively arid environments. - Establishing climate-smart technologies and practices to ensure sustainability in animal husbandry, dairying and fisheries sub- sectors - Ensuring sustainable land and forest management to reduce GHG emissions and grow forest-based economy in India
 Enhancing inclusiveness and technology	Improving gender inclusion - Ensuring inclusion of women farmers under all schemes in the agriculture and allied sectors - Empowering women farmers by making systemic investments to enhance their participation, productivity and creating sustainable agriculture-based livelihoods for rural women	Increasing role of technology Leveraging ICT tools and digital markets to address market asymmetry in agriculture and allied sectors using e-NAM - Boosting agri-tech development by using the India Digital Ecosystem of Agriculture (IDEA) framework and AgriStack - Promoting use of modern technologies – artificial intelligence, machine learning, robotics and blockchain – via the National e-Governance Plan in Agriculture (NeGP-A)
 Stakeholder collaborations	Increasing government investment in agriculture & allied infrastructure - Agriculture Infrastructure Fund mobilised INR 30,000 crore for post-harvest management infrastructure - Targeting the food processing industry (Pradhan Mantri Kisan Sampada Yojana) to reduce wastage of agricultural produce and minimise post-harvest losses - Strengthening meat processing and value addition infrastructure in India	Improving Public-Private Partnerships (PPP) - Leveraging PPP and international cooperation to strengthen agricultural research and technology transfer for better climate mitigation and adaptation - Effectively carry out genetic enhancement of plants/animals/fish for higher productivity - Promoting PPPs in farm mechanisation, advisory and post-harvest management

40 Government of India. (2023). *Demands for Grants of Central Government 2023-2024*.

Recommended Readings

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Joshi, P.K., Varshney, D. (2022). Agricultural Technologies in India: A Review. National Bank for Agricultural and Rural Development. NABARD Research and Policy Series No. 5/2022. <https://www.nabard.org/auth/writereaddata/tender/1507223612Paper-5-Agricultural-Tech-in-India-Dr.Joshi-&Varshney.pdf>

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सत्यमेव जयते

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